



Decision Trees

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What is a Decision Tree?



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- A **decision tree** is a flowchart-like model that splits data by feature tests and outputs a prediction at leaves.
- Internal nodes: *tests* on features (e.g., Outlook = Sunny?).
- Edges: outcomes of the tests (e.g., Yes/No, threshold left/right).
- Leaves: final *decision* — a class label (classification) or a numeric value (regression).
- Built *greedily* by choosing splits that best reduce impurity (e.g., entropy, Gini, variance).

Inference with Decision Trees



Classification Tree

- Traverse from root to a leaf by evaluating tests on input $\mathbf{x}$.
- The leaf stores a **class label** (e.g., Play = Yes/No) or a class distribution.
- Prediction can be the majority class at the leaf; confidence from class frequencies.

Regression Tree

- Same traversal, but the leaf stores a **numeric value** (e.g., mean of targets in that leaf).
- Prediction is typically the mean/median of training targets that fall into that leaf.

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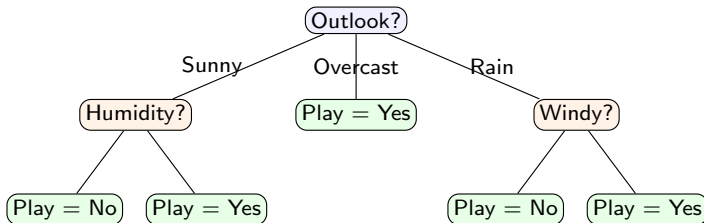
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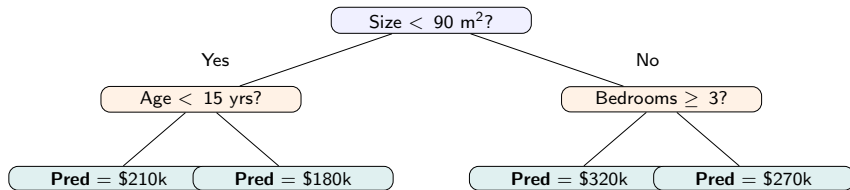
Simple Example (Play Tennis)

- Features: Outlook, Humidity, Windy.
- Target: $\text{Play} \in \{\text{Yes}, \text{No}\}$.
- Training instances are split by the *best* test at each node.
- Walk the path defined by a new input to get prediction at a leaf.



Simple Example (Regression Tree: House Price)

- Features: Size (m^2), Age (years), Bedrooms.
- Target: Price (in \$1,000s). Leaves store the **mean** of training targets in that region.
- Inference: traverse tests $\Rightarrow$ output numeric prediction at the leaf.



Notes: Split thresholds are chosen to minimize impurity for regression (e.g., MSE/variance). Leaves hold the mean target of training samples in that region (optionally the median).

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Why Decision Trees? (Quick Notes)



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- **Interpretability**: human-readable rules (paths from root to leaf).
- **Handles** categorical & numeric features; missing values can be treated with surrogates/imputation.
- **Fast inference**: simple test sequence $\Rightarrow$ leaf prediction.
- Basis for **powerful ensembles** (Random Forests, Gradient Boosted Trees).



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Definition

$$H(S) = - \sum_{i=1}^K p_i \log_2 p_i$$

- Measures the **uncertainty** (or impurity) of a dataset S .
- p_i : proportion of samples in class i .
- High entropy $\Rightarrow$ more mixed classes (uncertain).
- Low entropy $\Rightarrow$ purer class distribution.

Entropy: Example Dataset (Play Tennis)



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Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Target attribute: Play (Yes / No)

Entropy: Class Distribution



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- From dataset:

Yes = 9, No = 5

- Convert to probabilities:

$$p(\text{Yes}) = \frac{9}{14} \approx 0.64, \quad p(\text{No}) = \frac{5}{14} \approx 0.36$$

- These probabilities p_i will be plugged into the entropy formula.

Entropy: Calculation



Formula

$$H(S) = - \sum_{i=1}^K p_i \log_2 p_i$$

- Substituting class probabilities:

$$H(S) = - \left(0.64 \log_2 0.64 + 0.36 \log_2 0.36 \right)$$

- Result:

$$H(S) \approx 0.94$$

Interpretation: dataset is **impure**, since both classes appear significantly.

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Entropy: Pure Dataset Example

Sample	Outlook	Temp	Play
1	Sunny	Hot	Yes
2	Rain	Mild	Yes
3	Overcast	Cool	Yes
4	Sunny	Mild	Yes

- Class distribution: 100% Yes, 0% No.
- Entropy:

$$H(S) = -(1.0 \log_2 1.0 + 0) = 0$$

- Interpretation: Dataset is **pure**, no uncertainty.



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Entropy: Nearly Pure Dataset



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- Suppose dataset has 10 samples:

$$\text{Yes} = 9, \quad \text{No} = 1$$

- Probabilities:

$$p(\text{Yes}) = 0.9, \quad p(\text{No}) = 0.1$$

- Entropy:

$$H(S) = -(0.9 \log_2 0.9 + 0.1 \log_2 0.1) \approx 0.47$$

- Interpretation: Mostly pure, small uncertainty due to one minority class.

Information Gain



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Definition

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

- S : dataset; A : attribute used to split.
- S_v : subset of S where attribute $A = v$.
- Measures the **reduction in entropy** after splitting by A .
- Attribute with highest IG is chosen for splitting (ID3 algorithm).

Information Gain: Step 1

- Full dataset S (PlayTennis): 14 samples.
- Class distribution:

$$\text{Yes} = 9, \quad \text{No} = 5$$

- Entropy of S :

$$H(S) \approx 0.94$$

- This is the baseline impurity before any split.



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Information Gain: Step 2 (Split on Outlook)

- Attribute $A = \text{Outlook}$ has 3 values: Sunny, Overcast, Rain.
- Subsets:

S_{Sunny} : 5 samples (2 Yes, 3 No)

S_{Overcast} : 4 samples (4 Yes, 0 No)

S_{Rain} : 5 samples (3 Yes, 2 No)



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Information Gain: Step 3 (Entropy of Subsets)



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- $H(S_{\text{Sunny}}) = -(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5}) \approx 0.97$
- $H(S_{\text{Overcast}}) = -(1.0 \log_2 1.0 + 0) = 0$
- $H(S_{\text{Rain}}) = -(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5}) \approx 0.97$

Information Gain: Step 4 (Result)



Formula

$$IG(S, Outlook) = H(S) - \sum_{v \in \{Sunny, Overcast, Rain\}} \frac{|S_v|}{|S|} H(S_v)$$

- Weighted average entropy:

$$\frac{5}{14} \cdot 0.97 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.97 \approx 0.69$$

- Information Gain:

$$IG(S, Outlook) \approx 0.94 - 0.69 = 0.25$$

- Outlook provides reduction in uncertainty $\Rightarrow$ a good splitting attribute.

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Information Gain: Attribute Splits



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Summary

- For each candidate attribute A (Outlook, Humidity, Wind, Temperature):
 - ① Partition the dataset by values of A .
 - ② Compute entropy $H(S_v)$ for each subset S_v .
 - ③ Compute weighted average entropy and subtract from $H(S)$.
- This gives $IG(S, A)$ — the reduction in uncertainty due to attribute A .
- **Exercise:** Complete detailed entropy and IG calculations for Humidity, Wind, and Temperature using the PlayTennis dataset.

Information Gain: Comparison

Attribute	$IG(S, \text{Attribute})$
Outlook	0.25
Humidity	0.15
Wind	0.05
Temperature	0.03

Conclusion: Attribute with highest IG is **Outlook**, chosen as the root node (ID3).



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Definition

$$Gini(S) = 1 - \sum_{i=1}^K p_i^2$$

- Another measure of impurity used in CART.
- p_i : proportion of samples in class i .
- $Gini(S) = 0$ when all samples belong to one class (pure).
- Larger Gini $\Rightarrow$ higher impurity.

Gini Index: Example (PlayTennis Dataset)



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Summary

- Dataset S : 14 samples

- Class distribution:

$$\text{Yes} = 9, \quad \text{No} = 5$$

- Probabilities:

$$p(\text{Yes}) = \frac{9}{14} \approx 0.64, \quad p(\text{No}) = \frac{5}{14} \approx 0.36$$

- Gini Index:

$$\text{Gini}(S) = 1 - (0.64^2 + 0.36^2) \approx 0.46$$

Interpretation: moderate impurity, dataset is mixed but not balanced.

Gini Index: Pure Dataset



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- Suppose dataset has all samples from one class:

$$p(\text{Yes}) = 1.0, \quad p(\text{No}) = 0.0$$

- Gini Index:

$$\text{Gini}(S) = 1 - (1.0^2 + 0.0^2) = 0$$

- Interpretation: dataset is **pure** (no uncertainty).

Gini Index: Balanced Dataset



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Summary

- Suppose dataset has equal classes:

$$p(\text{Yes}) = 0.5, \quad p(\text{No}) = 0.5$$

- Gini Index:

$$\text{Gini}(S) = 1 - (0.5^2 + 0.5^2) = 0.5$$

- Interpretation: maximum impurity for 2-class case (most uncertain).

Comparison of Impurity Measures

- **Entropy**: logarithmic measure of disorder (ID3).
- **Information Gain**: uses entropy difference to choose splits.
- **Gini Index**: simpler computation, used in CART.
- In practice: Entropy and Gini often lead to similar trees.



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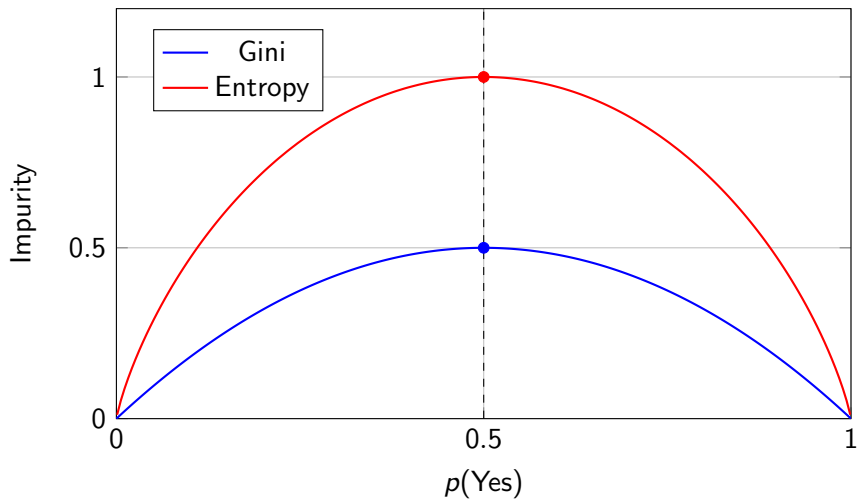
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Comparison: Entropy vs Gini (Binary Case)



Observation: Both reach max impurity at $p = 0.5$, but scales differ: Entropy = 1, Gini = 0.5.



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Summary

- Builds a decision tree using **Entropy** and **Information Gain**.
- **Greedy approach**: at each node, choose the attribute with the highest *IG*.
- Recursive splitting until:
 - All samples in node belong to the same class, or
 - No attributes remain (majority class is chosen).

ID3 Example Dataset (Play Tennis)



Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Target: Play (Yes / No)

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Step 1: Compute Entropy of Dataset

- Class distribution:

Yes = 9, No = 5

- Entropy of dataset S :

$$H(S) = - \left(\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right) \approx 0.94$$

- This is the baseline impurity.



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Step 2: Compute Information Gain

- Candidate attributes: Outlook, Humidity, Wind, Temperature.
- Compute IG for each by splitting dataset:
 - $IG(S, Outlook) \approx 0.25$
 - $IG(S, Humidity) \approx 0.15$
 - $IG(S, Wind) \approx 0.05$
 - $IG(S, Temperature) \approx 0.03$
- Attribute with highest IG: **Outlook**.



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Step 3: Split on Best Attribute

- Root node: **Outlook**.
- Branches:
 - Outlook = Overcast $\Rightarrow$ Play = Yes (pure).
 - Outlook = Rain $\Rightarrow$ split further on Wind.
 - Outlook = Sunny $\Rightarrow$ split further on Humidity.
- Recursively apply ID3 to remaining subsets.



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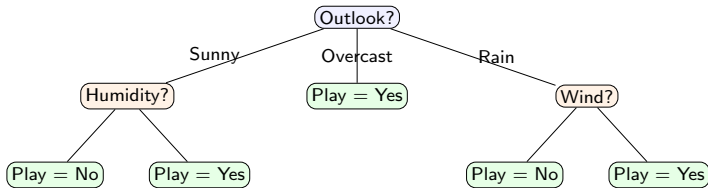
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Resulting Decision Tree (ID3 on Play Tennis)



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Summary

- Uses **Entropy** and **Information Gain**.
- Greedy recursive splitting to reduce impurity.
- Works well on categorical attributes (e.g. Outlook, Wind).
- Limitations:
 - Tends to overfit if tree grows too deep.
 - Biased toward attributes with many values.
- Extensions: C4.5 addresses these issues.

C4.5 Algorithm



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Summary

- Extension of ID3 to overcome its limitations.
- Handles both **categorical** and **continuous** attributes.
- Uses **Gain Ratio** instead of pure Information Gain.
- Includes **pruning** to avoid overfitting.

C4.5 Improvements over ID3



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Summary

- **Bias issue in ID3:** IG favors attributes with many values.
- **Solution: Gain Ratio**

$$GR(S, A) = \frac{IG(S, A)}{SplitInfo(S, A)}, \quad SplitInfo(S, A) = - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$$

→ **Choose the attribute with the highest Gain Ratio.**

- **Continuous attributes:** find threshold splits (e.g., Temp $\leq$ 75).
- **Pruning:** remove branches with little predictive power.

Example: Gain Ratio

- Attribute: Outlook
- Information Gain: $IG(S, Outlook) \approx 0.25$
- SplitInfo:

$$-\left(\frac{5}{14} \log_2 \frac{5}{14} + \frac{4}{14} \log_2 \frac{4}{14} + \frac{5}{14} \log_2 \frac{5}{14}\right) \approx 1.58$$

- Gain Ratio:

$$GR(S, Outlook) = \frac{0.25}{1.58} \approx 0.16$$

- **Interpretation:** Normalizes IG, avoids bias toward multi-valued attributes.



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Handling Continuous Attributes



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Summary

- ID3 cannot handle continuous (numeric) attributes directly.
- C4.5 extends ID3 by introducing **threshold splits**.
- Idea: choose a threshold T^* , then split into

$$\{x \mid A \leq T^*\}, \quad \{x \mid A > T^*\}.$$

- Threshold T^* is selected to maximize **Gain Ratio**.

Continuous Attributes: Step 1



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Summary

- Attribute: Temperature (numeric).
- Values in dataset (sorted): 64, 65, 68, 69, 70, 71, 72, 75, 80, 81, 83, 85, 86, 90.
- Possible thresholds: midpoints between consecutive values.

$$T^* \in \{64.5, 66.5, 68.5, 69.5, \dots, 87.5\}.$$

- Each candidate threshold yields a binary split of the dataset.

Continuous Attributes: Step 2



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- Example candidate: $T^* = 75$.

- Split:

$$S_1 = \{\text{Temp} \leq 75\}, \quad S_2 = \{\text{Temp} > 75\}.$$

- Distribution:

- S_1 : 8 samples (6 Yes, 2 No)
- S_2 : 6 samples (3 Yes, 3 No)

- Compute entropy for each subset.

Continuous Attributes: Step 3



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- $H(S_1) = -(\frac{6}{8} \log_2 \frac{6}{8} + \frac{2}{8} \log_2 \frac{2}{8}) \approx 0.81$

- $H(S_2) = -(\frac{3}{6} \log_2 \frac{3}{6} + \frac{3}{6} \log_2 \frac{3}{6}) = 1.0$

- Weighted entropy:

$$\frac{8}{14} \cdot 0.81 + \frac{6}{14} \cdot 1.0 \approx 0.90$$

- Information Gain:

$$IG(S, T^* = 75) = H(S) - 0.90 = 0.94 - 0.90 = 0.04$$

Continuous Attributes: Step 4



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Summary

- Repeat calculation for all candidate thresholds.
- Choose threshold T^* with highest Gain Ratio.
- In practice:
 - Thresholds near “natural splits” in the data work best.
 - For Temperature, a split around 75 may separate many Play=Yes from Play=No.
- **Conclusion:** C4.5 can handle continuous attributes by converting them into binary splits.

Pruning in C4.5



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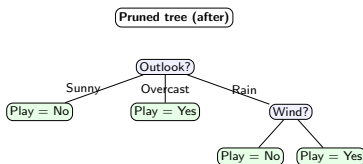
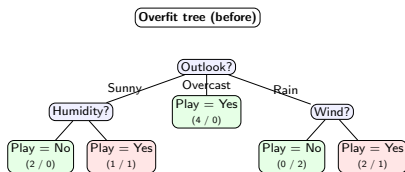
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Summary

- Motivation: ID3 tends to overfit with deep trees.
- **Pre-pruning:** stop splitting if node has too few samples or no significant IG.
- **Post-pruning:** build full tree, then remove branches that do not improve validation accuracy.
- **Intuition:** pruning = “simplify the tree to generalize better”.



Note: Tiny, low-support leaves (red) often capture noise; pruning collapses them into simpler, more general leaves (green), improving validation performance.

Pruning in C4.5: Why and What



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Summary

- **Why:** Deep trees fit training noise $\Rightarrow$ poor generalization (overfitting).
- **What:** Remove weak branches and replace subtrees by simpler leaves.
- Two strategies:
 - **Pre-pruning** (early stopping): stop splitting under certain conditions.
 - **Post-pruning** (C4.5 default): grow full tree, then prune bottom-up.

Pre-pruning: When to Stop and What to Predict



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Typical stopping rules (examples)

- Node has **low support**: $|S| < N_{\min}$.
- No split yields **significant** impurity reduction (IG/GR below threshold).
- Node is **nearly pure**: majority proportion $\geq \tau$ (e.g., $\tau = 0.95$).

Leaf prediction at a stopped node

- **Classification**: predict **majority class** at the node; report class distribution as confidence.
- **Regression**: predict **mean** (or median) of targets in the node; report variance as uncertainty.
- Tie-breaking: prefer the class with larger prior or use a fixed order.

Post-pruning (C4.5): Local Replacement



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- Traverse the tree **bottom-up**. For each internal node t :
 - ① Compute **subtree error** at t : sum of leaf errors under t .
 - ② Compute **leaf error** if t were replaced by a single leaf (majority class / mean).
 - ③ **Prune** if the estimated leaf error is not worse than subtree error (within a tolerance).
- C4.5 uses a **pessimistic error estimate** (binomial correction) to compare fairly.

Intuition: If a subtree does not reduce the estimated error enough, replace it with a simpler leaf.

Classification Error for Pruning (Intuition)



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- At any node t with N samples and class counts $\{n_c\}$:
 - Leaf error** if pruned to a single leaf:

$$\text{err}_{\text{leaf}}(t) = N - \max_c n_c$$

- Subtree error** (before pruning): sum of misclassifications across leaves below t :

$$\text{err}_{\text{subtree}}(t) = \sum_{\ell \in \text{leaves}(t)} (N_{\ell} - \max_c n_{c,\ell})$$

- Decision:** prune if $\widehat{\text{err}}_{\text{leaf}}(t) \leq \widehat{\text{err}}_{\text{subtree}}(t)$ (using pessimistic estimates).

C4.5 Pessimistic Error (Sketch)



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- Let e be observed errors on N samples at a node/leaf, $\hat{p} = e/N$.
- Estimate an **upper-bound** error rate using binomial standard error:

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}, \quad \hat{p}_{\text{upper}} \approx \hat{p} + z \cdot SE$$

- Compare $\hat{p}_{\text{upper}}$ of the subtree vs. the single-leaf alternative (majority).
- If subtree is not significantly better, **prune**.

Note: Quinlan's C4.5 uses a specific continuity-corrected estimate; this slide gives the core idea for class use.

Worked Example: Should We Prune This Subtree?



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- Node t has $N = 20$ samples: **Yes**=13, **No**=7.
- If pruned to a single leaf (majority=Yes):

$$\text{err}_{\text{leaf}} = 20 - 13 = 7$$

- Current subtree has two leaves:
 - Left leaf: 8 Yes, 2 No $\Rightarrow$ errors = 2
 - Right leaf: 5 Yes, 5 No $\Rightarrow$ errors = 5

$$\text{err}_{\text{subtree}} = 2 + 5 = 7$$

- **Tie:** Without correction, equal errors. With pessimistic correction, subtree often loses $\Rightarrow$ **prune**.

Take-away: If detailed splits don't clearly reduce estimated error, collapse to a majority leaf.

Leaf Prediction: After Pruning or Early Stop



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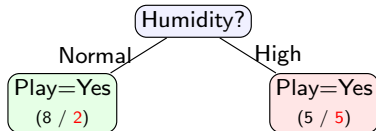
Missing Values

Summary

- **Classification:**
 - Predict **majority class** at the node.
 - Optionally output class probabilities from class frequencies at the node.
- **Regression:**
 - Predict **mean** (or median) target at the node.
 - Report the node variance as an uncertainty indicator.
- **During inference:** for a new sample, *traverse* tests $\Rightarrow$ leaf; return the leaf's prediction.

Local Pruning Effect (Visual)

Before



Pruning collapses an uncertain split into a single majority leaf.

After (Pruned)

Play=Yes
Total=20, Errors=7



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Pruning: Cheat Sheet



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Summary

- **Pre-pruning (stop early):**
 - Low support, tiny IG/GR, or near-pure node $\Rightarrow$ stop.
 - Predict majority (classification) / mean (regression).
- **Post-pruning (C4.5):**
 - ① Compute subtree vs. leaf error at each internal node (bottom-up).
 - ② Use pessimistic error to account for sampling noise.
 - ③ If subtree not significantly better $\Rightarrow$ **replace by leaf**.
- **Inference rule (always):** traverse to a leaf, return the leaf's prediction and (optionally) confidence.

C4.5 Summary

- Improves ID3 by using Gain Ratio.
- Can handle continuous and missing values.
- Includes pruning to reduce overfitting.
- Widely used, forms the basis of popular implementations (e.g., Weka J48).



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CART: Overview



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Summary

- **CART** builds binary trees for both **classification** and **regression**.
- **Classification**: choose splits that reduce **Gini impurity**.
- **Regression**: choose splits that reduce **variance** / **SSE** (sum of squared errors).
- **Always binary splits**: continuous and categorical attributes are handled via binary partitions.
- **Pruning**: cost-complexity pruning with hyperparameter α .

CART for Classification: Gini Impurity



Node impurity

$$G(t) = 1 - \sum_{k=1}^K p(k | t)^2$$

where $p(k | t)$ is the class proportion at node t .

Split quality at node t

For a binary split s producing left/right children t_L, t_R with sizes N_L, N_R :

$$\Delta G(s, t) = G(t) - \frac{N_L}{N_t} G(t_L) - \frac{N_R}{N_t} G(t_R)$$

- **Greedy choice:** pick $s^* = \arg \max_s \Delta G(s, t)$.
- Repeat recursively until a stopping rule is met.

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CART for Regression: Variance/SSE Reduction



Node impurity (SSE form)

$$I(t) = \sum_{i \in t} (y_i - \bar{y}_t)^2$$

where $\bar{y}_t$ is the mean target in node t .

Split quality

$$\Delta I(s, t) = I(t) - I(t_L) - I(t_R)$$

- **Greedy choice:** pick $s^* = \arg \max_s \Delta I(s, t)$.
- Leaf prediction: $\hat{y}(\text{leaf}) = \bar{y}_{\text{leaf}}$ (or median).

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Handling Continuous Attributes



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Summary

- **Binary thresholds** for numeric A : sort unique values, consider midpoints.
- For each candidate threshold $A \leq T^*$ vs. $A > T^*$:
 - **Classification**: evaluate $\Delta G(s, t)$.
 - **Regression**: evaluate $\Delta I(s, t)$ (SSE reduction).
- Choose the threshold with the **largest impurity reduction**.

Handling Categorical Attributes (as Binary Splits)



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Summary

- For a categorical attribute with values $\{v_1, \dots, v_m\}$, CART finds a **binary partition**:

$$\{v_{i_1}, \dots, v_{i_r}\} \text{ vs. } \{v_{j_1}, \dots, v_{j_{m-r}}\}.$$

- Search over candidate groupings (heuristics used if m is large).
- Evaluate impurity reduction (Gini or SSE) and pick the best binary partition.

Cost-Complexity Pruning (CART)



Objective for a tree T

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

- $R(T)$: empirical error of T (misclassification rate or SSE aggregated over leaves).
- $|T|$: number of leaves; $\alpha \geq 0$: complexity penalty.

- **Weakest-link pruning**: iteratively prune the internal node t with smallest

$$g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1},$$

where T_t is the subtree rooted at t .

- This produces a **sequence of nested subtrees**. Pick α by cross-validation.

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Choosing α (Model Selection)



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Summary

- Generate a subtree sequence via weakest-link pruning.
- Use **cross-validation** to estimate $R_\alpha(T)$ on held-out folds.
- Select α^* (and the subtree) with minimal estimated generalization error.
- **One-SE rule** (optional): pick the simplest subtree within one standard error of the best.

Missing Values and Surrogate Splits

- If the primary split feature is missing for a sample, route it using a **surrogate split** (another feature whose split best agrees with the primary on training data).
- Surrogates are ranked during training by their agreement with the primary split.
- Benefit: more robust routing at inference time when some features are missing.



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CART Classification: Toy Tree



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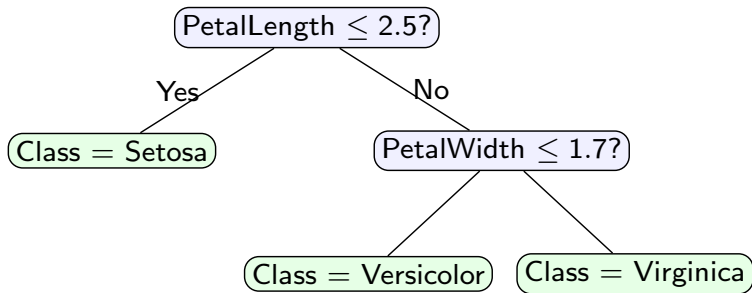
ID3 Algorithm

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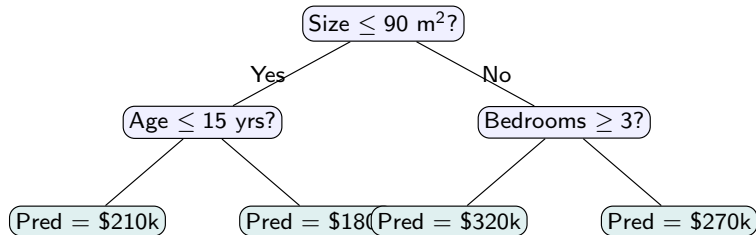
Summary



Greedy splits

chosen by maximum Gini reduction; leaves predict majority class.

CART Regression: Toy Tree



splits chosen by maximum SSE reduction; leaves predict mean target in the leaf.



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CART vs ID3/C4.5 (At a Glance)



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Summary

- **Splits:** CART uses **binary** splits; ID3/C4.5 can produce multiway splits.
- **Criteria:** Gini (classification) & SSE (regression) vs. Entropy/IG (ID3), Gain Ratio (C4.5).
- **Pruning:** CART uses **cost-complexity** pruning; C4.5 uses error-based (pessimistic) pruning.
- **Outputs:** class label or probability (classification); mean/median (regression).

CART: Summary



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Summary

- Unified framework for **classification** (Gini) and **regression** (SSE).
- **Binary** splits on both continuous and categorical attributes.
- **Cost-complexity pruning** produces a subtree sequence; select via cross-validation.
- Robust handling of missing values via **surrogate splits**.
- Strong baseline and foundation for tree ensembles (Random Forests, Gradient Boosted Trees).



Missing Values

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Missing Values: Challenges



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- In real-world data, some feature values are missing (e.g., Humidity = ?).
- Naïve strategies:
 - Drop records with missing values (wasteful).
 - Impute with mean/median/mode (may bias).
- Decision trees offer **better strategies**:
 - Use available features to still route the sample.
 - Employ **surrogate splits**.

Surrogate Splits (CART)



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Summary

- **Idea:** Find alternative features that best mimic the primary split.
- Example:
 - Primary split: Outlook $\rightarrow$ Sunny, Rain.
 - Surrogate: Humidity often correlates with Outlook.
- During training:
 - Rank features by agreement with the primary split.
 - Store top- k surrogates.
- During prediction:
 - If primary attribute is missing, use surrogate to decide branch.

Missing Values During Training



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Summary

- **C4.5:**

- For a candidate split on feature A , compute IG / Gain Ratio using only samples with observed A ; *fractionally weight* missing- A samples across branches according to observed branch frequencies.
- After selecting the split, propagate missing- A samples to children with those fractional weights (so they still contribute to counts).

- **CART:**

- Evaluate impurity reduction using samples with observed A only.
- Learn **surrogate splits**: rank other features by *agreement* with the primary split (measured on cases where both features are observed), and store top- k surrogates at the node.
- When a training sample is missing on the primary feature, route it using the best available surrogate; fall back to next surrogates if needed.

- **ID3**: no built-in handling; requires preprocessing (imputation or dropping).

Missing Values: Practical Notes



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Summary

- Surrogate splits make trees **robust** to missing data.
- C4.5 vs CART:
 - C4.5 distributes a sample fractionally across branches (based on observed frequencies).
 - CART uses surrogate splits for hard routing.
- Common libraries (scikit-learn) often require preprocessing (imputation), while `rpart` (R) implements surrogate splits.

Summary



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Decision Trees: Summary (1/2)



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- **Concept:**

- Non-parametric models for classification & regression.
- Inference = traverse tree until a leaf; prediction = class label or numeric value.

- **Impurity measures:**

- Entropy & Information Gain (ID3).
- Gain Ratio (C4.5).
- Gini Index, Variance reduction (CART).

- **Algorithms:**

- ID3: splits by max Information Gain.
- C4.5: Gain Ratio, continuous attributes, pruning, fractional assignment.
- CART: always binary splits, Gini/SSE, cost-complexity pruning, surrogate splits.

Decision Trees: Summary (2/2)



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- **Practical issues:**
 - Handle continuous attributes via thresholding.
 - Missing values: C4.5 (fractional), CART (surrogates).
 - Overfitting $\Rightarrow$ pruning.
- **Key insight:** Decision trees are simple, interpretable, and form the foundation for powerful ensemble methods:
 - Random Forests
 - Gradient Boosted Trees